

EGR 345 HW 2: Impulse and Step Response

For each transfer function below, find the impulse response and the step response.

Verify your answer by overlaying your symbolic result with results from the appropriate python functions.

1. $G(s) = \frac{1}{s^2 + s + 49}$

2. $G(s) = \frac{1}{s^2 + 20s + 49}$

3. $G(s) = \frac{1}{(s + 1)(s^2 + s + 49)}$